SEQUENCES AND SUMMATIONS

Subtitle

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OUTLINE

Sequence

Definition 1

Ordered list of elements created by a function to the set S.
Notation: a_n represents n^{th} term.

Example

If $a_n = 2n$, find $\{a_n\}$.

$$\begin{aligned} a_0 &= 2(0) = 0 \\ a_1 &= 2(1) = 2 \\ a_2 &= 2(2) = 4 \\ a_2 &= 2(3) = 6 \end{aligned} \tag{1}$$

$\therefore \{a_n\} = \{0, 2, 4, 6, ...\}$
A sequence can have finite or infinite terms.

Arithmetic Sequence (Progression)

A sequence formed by adding the initial term, a, and the product of the common difference, d, and the term number, n.

$\{a_n\} = a, a+d, a+2d, a+3d, ..., a+nd$ ...
The *initial term* a and the *common difference* d are real numbers.
Remark: An arithmetic progression is a discrete analogue of the linear function $f(x) = dx + a$.

Example

Let $a = 5$ and $d = 2$. Find the first 5 terms of $\{a_n\}$.
Sol $a_n = 5 + 2n$

Example

$\{s_n\}$ where $s_n = -1 + 4n$ is an arithmetic progression with initial term -1 and common difference 4.

Example

Find a and d for the sequence $\{7, 4, 1, -2, ...\}$

Geometric Sequence

A sequence formed by multiplying the initial term, a, by the common ration to the n^{th} power, r.

The *initial term* a and the *common ratio* r are real numbers.

$\{a_n\} = a, ar, ar^2, ar^3, \cdot.., ar^n$

Remark: A geometric progression is a discrete analogue of the exponential function $f(x) = ar^x$.

Example

Let $a = 4$ and $r = 3$. Find the first 5 terms of the geometric progression $\{a_n\}$.

Sol $a_n = 4(3)^n$, and $\{a_n\} = \{4, 12, 36, 108, 324, ...\}$

Example

Find a and r for the geometric sequence $\{a_n\} = \{3, \frac{3}{2}, \frac{3}{4}, \frac{3}{8}, ...\}$.

$a_n = ar^n = 3r^n$, $r = \frac{1}{2}$

Recurrence Relation
An **equation** that expresses a_n in terms of one or more of the previous terms of the sequence. The **Initial conditions**, that specify the terms that precede the first term, are required to specify terms that precede the first term where the relation takes effect.

Example
$a_n = a_{n-1} + 2a_{n-2}$ for $n \geq 2$, where $a_0 = 2$, and $a_1 = 5$.
$a_2 = a_1 + 2a_0 = 5 + 2(2) = 9$
$a_3 = a_2 + 2a_1 = 9 + 2(5) = 19$
$a_4 = a_3 + 2a_2 = 19 + 2(9) = 37$
$a_n = \{2, 5, 9, 19, 37, ...\}$

Fibonacci Sequence
The Fibonacci sequence, f_0, f_1, f_2, ... , is defined by the initial conditions $f_0 = 0$, $f_1 = 1$, and the recurrence relation
$f_n = f_{n-1} + f_{n-2}$
for $n = 2, 3, 4, ...$

Example Find the Fibonacci numbers f_2, f_3, f_4, f_5, and f_6.
Initial conditions $f_0 = 0$, $f_1 = 1$

$$f_2 = f_1 + f_0 = 1 + 0 = 1$$
$$f_3 = f_2 + f_1 = 1 + 1 = 1$$
$$f_4 = f_3 + f_2 = 2 + 1 = 3$$
$$f_5 = f_4 + f_3 = 3 + 2 = 5$$
$$f_6 = f_5 + f_4 = 5 + 3 = 8$$

Solving Recurrence Relations

Finding a non-recursive formula to calculate a_n is called solving the recurrence relation. The solution is called the **closed formula**.

Example

Let a_n be the sequence that satisfies the recurrence relation $a_n = a_{a-1} + 3$ for $n \geq 1$ with $a_0 = 2$.

Using the method for iteration, involving substitution,

$a_0 = 2$
$a_1 = 2 + 3$
$a_2 = (2 + 3) + 3 = 2 + 2(3)$
$a_3 = (2 + 3 + 3) + 3 = 2 + 3(3)$
$a_4 = 2 + 4(3)$

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$a_n = 2 + 3n$

Some Useful Sequences.	
nth Term	***First 10 Terms***
n^2	1, 4, 9, 16, 25, 36, 49, 64, 81, 100, …
n^3	1, 8, 27, 64, 125, 216, 343, 512, 729, 1000, …
n^4	1, 16, 81, 256, 625, 1296, 2401, 4096, 6561, 10000, …
f_n	1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, …
2^n	2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, …
3^n	3, 9, 27, 81, 243, 729, 2187, 6561, 19683, 59049, …
$n!$	1, 2, 6, 24, 120, 720, 5040, 40320, 362880, 3628800, …

Example Find formulae for the sequences with the following first five terms:
(a) 1, $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{16}$
(b) 1, 3, 5, 7, 9
(c) 1, -1, 1, -1, 1.

Solution:
(a) We recognize that the denominators are powers of 2. The sequence with $a_n = \frac{1}{2^n}$, $n = 0, 1, 2, ...$ is a possible match. This proposed sequence is a geometric progression with $a = 1$ and $r = \frac{1}{2}$.
(b) We note that each term is obtained by adding 2 to the previous term. The sequence with $a_n = 2n + 1$, $n = 0, 1, 2, ...$ is a possible match. This proposed sequence is an arithmetic progression with $a = 1$ and $d = 2$.
(c) The terms alternate between 1 and -1. The sequence with $a_n = (-1)^n$, $n = 0, 1, 2, ...$ is a possible match. This proposed sequence is a geometric progression with $a = 1$ and $r = -1$.

Sigma Notation

To express the sum of the terms of the sequence $a_n = \{a_m, a_{m+1}, \cdots, a_m\}$, we write

$$\sum_{i=m}^{n} a_i$$

which represents $\{a_m + a_{m+1} + a_{m+1} + \cdots + a_n\}$

Example

Use sigma notation to express the sum of the first 100 terms of the sequence a_i where $a_i = \frac{1}{i}$ for $i = 1, 2, \cdots$

$a_1 = \frac{1}{1}$, $a_2 = \frac{1}{2}$, $a_3 = \frac{1}{3} \cdots$

$a_i = \frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{100}$

$$\sum_{i=1}^{100} \frac{1}{i} = \frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{100}$$

Example What is the value of $\sum_{i=5}^{9} i^2$?
Solution

$$\begin{aligned}\sum_{i=5}^{9} i^2 &= 5^2 + 6^2 + 7^2 + 8^2 + 9^2 \\ &= 25 + 36 + 49 + 64 + 81 \\ &= 255\end{aligned}$$

Example What is the value of $\sum_{i=7}^{10} (-1)^i$?
Solution

$$\begin{aligned}\sum_{i=7}^{10} (-1)^i &= (-1)^7 + (-1)^8 + (-1)^9 = (-1)^{10} \\ &= -1 + 1 - 1 + 1 \\ &= 0\end{aligned}$$

Change of index

Suppose we have the sum

$$\sum_{j=1}^{5} j^2$$

but want the index of summation to run between 0 and 4 rather than from 1 to 5. To do this, we let $k = j - 1$. Then the new summation index runs from 0 (because $k = 1 - 1 = 0$ when $j = 1$) to 4 (because $k = 5 - 1 = 4$ when $j = 5$), and the term j^2 becomes $(k+1)^2$. Hence,

$$\sum_{j=1}^{5} j^2 = \sum_{k=0}^{4} (k+1)^2$$

Basic Properties

$$\sum_{k=1}^{n} c a_k = c \sum_{k=1}^{n} a_k$$

$$\sum_{k=1}^{n} c = cn$$

$$\sum_{k=1}^{n} (a_k \pm b_k) = \sum_{k=1}^{n} a_k \pm \sum_{k=1}^{n} b_k$$

$$\sum_{k=1}^{n} a_k = \sum_{k=1}^{j} a_k + \sum_{k=j+1}^{n} a_k \qquad \text{for } i \leq j < j+1 < n$$

$$\sum_{k=1}^{n} a_k = \sum_{k=1+m}^{n+m} a_{k-m}$$

Useful Summation Formula

$$\sum_{i=1}^{n} i = \frac{(n)(n+1)}{2}$$

Example $$\sum_{i=1}^{7} i = \frac{(7)(8)}{2} = 28$$

$$\sum_{i=1}^{n} i^2 = \frac{(n)(n+1)(2n+1)}{6}$$

Example $$\sum_{i=1}^{7} i^2 = \frac{(7)(8)(2(7)+1)}{6} = 140$$

$$\sum_{i=1}^{n} i^3 = \frac{(n^2)(n+1)^2}{4}$$

Example $$\sum_{i=1}^{7} i^3 = \frac{(7^2)(8^2)}{4} = 784$$

$$\sum_{k=0}^{n} ar^k = \frac{(ar^{n+1}-1)}{r-1}, r \neq 1$$

Example $$\sum_{k=0}^{7} (2.3^k + 5.2^k) \quad = \sum_{k=0}^{7} 2.3^k + \sum_{k=0}^{7} 5.2^k$$

Example - cont

Example $$\sum_{k=0}^{7}(2.3^k + 5.2^k) = \sum_{k=0}^{7} 2.3^k + \sum_{k=0}^{7} 5.2^k$$
$$= \frac{2.3^8 - 2}{2} + \frac{5.2^8 - 5}{1}$$
$$= \frac{13122 - 2}{2} + \frac{1275}{1}$$
$$= 7835$$

The End